Variables Listing

Questionnaire - internal data

Apart from your questions, you will find in the data record additional variables, as far as you have not deactivated this option when downloading the data record.

CASE Serial number of the subject
REF Reference, if such has been provided in the link to the questionnaire
LASTPAGE Page number of the questionnaire that has been edited and sent last
QUESTNNR ID of the questionnaire that has been edited
MODE Information if the questionnaire has been started by pretest or by a project member
STARTED Time the interviewee opened the questionnaire
FINISHED Information if the questionnaire has been completed up to the last page
TIME_001... Time that an interviewee has spend on a questionnaire page

Please note that you can not read the internal variables of the questionnaire with the function value(). For interview number and reference there are the PHP functions PHP function caseNumber() and PHP function reference()

Details about additional variables can be find in the manual: Additional Variables in the Data Set

Section AU: Actual Understanding

[AU01] Text Input
How predictions are made
"Based only on your experience of completing the previous task and the information that has been provided to ..."

AU01_01 [01]
Free text

[AU02] Multiple Choice
Most challenging classes
"Can you select the 3 categories that were the most challenging for the satellite image classifier to recognize?"

AU02 Most challenging classes: Residual option (negative) or number of selected options
Integer
AU02_01 Agricultural area
AU02_02 Forest
AU02_03 Industrial or commercial area
AU02_04 Railway
AU02_05 Residential area
AU02_06 Road
AU02_07 Public square or park
AU02_08 Waterbody
1 = Not checked
2 = Checked

[AU03] Multiple Choice
Easiest class
"Can you select the 3 categories that were the easiest for the satellite image classifier to recognize?"

AU03 Easiest class: Residual option (negative) or number of selected options
Integer
AU03_01 Agricultural area
AU03_02 Forest
AU03_03 Industrial or commercial area
AU03_04 Railway
AU03_05 Residential area
AU03_06 Road
AU03_07 Public square or park
AU03_08 Waterbody
1 = Not checked
2 = Checked

[AU04] Multiple Choice
Sorted confidence
"Imagine that you have sorted the following test set in a descending confidence order (from highest to lowest..."

AU04 Sorted confidence: Residual option (negative) or number of selected options
Integer

AU04_01 The model is more confident in its prediction for image 1 than for its prediction for image 4.

AU04_02 Image 1 is more likely to be correctly classified by the model than image 4.

AU04_03 The model is less confident in its prediction for image 4 than for its prediction for image 1, which suggests that the prediction for image 4 is likely to be incorrect.

AU04_04 I don't know.
1 = Not checked
2 = Checked

[AU05] Selection
Confident prediction
"Which of the following images is most likely to have the lowest confidence in the prediction?"

AU05 Confident prediction
1 = Image 1
2 = Image 2
3 = Image 3
4 = I don't know
-9 = Not answered

[AU06] Selection
Correct coverage plot
"Imagine that you have built a test set with the following predictions."

AU06 Correct coverage plot
1 = Test Coverage 1
2 = Test Coverage 2
3 = I don't know
-9 = Not answered

[AU07] Selection
Highest test coverage
"Consider the following test coverage plot to answer the question. Which category has the highest test coverage?"

AU07 Highest test coverage
1 = Bridge
2 = Building
3 = Field
4 = Forest
5 = Railway
6 = Road
7 = River
8 = Square
10 = I don't know
-9 = Not answered

[AU08] Selection
Road correctly classified
"Consider the following bar plot to answer the question. How many instances of "Road" were correctly classified?"

AU08 Road correctly classified
1 = 1
2 = 2
3 = 3
4 = 4
5 = I don't know
-9 = Not answered

[AU09] Selection
Algorithmic output
"What do you think would be the model's prediction for the following image?"

AU09 Algorithmic output
1 = Road
2 = Agricultural area
3 = Square or park
4 = Forest
5 = I don't know
-9 = Not answered

Section ID: Participant ID

[ID01] Text Input
Participant ID
"Please hold on for a moment while we enter your participant ID."

ID01_01 ID
Free text

[ID02] Internal Variables
userID test

ID02_02 userID
Text/string

Section PK: Prior Knowledge

[PK01] Text Input
Age
"How old are you?"

PK01_01 [01]
Free input (integer)

[PK02] Selection
Gender
"Which gender do you identify with?"

PK02 Gender
1 = Female
2 = Male
3 = Other
4 = Prefer not to say
-9 = Not answered

[PK03] Selection
Major
"Did you major in Computer Science or a related field (e.g., Data Science, Artificial Intelligence, etc.)?"

PK03 Major
1 = Yes
2 = No
-9 = Not answered

[PK04] Selection
Class
"Have you taken any class related to machine learning in your studies?"

PK04 Class
1 = Yes
2 = No
-9 = Not answered

[PK05] Selection
Train or test experience
"Have you worked on training or testing machine learning models before?"

PK05 Train or test experience
1 = Yes
2 = No
-9 = Not answered

[PK06] Selection
End-to-end projects
"Have you completed any machine learning projects end-to-end (from data collection to model deployment)?"

PK06 End-to-end projects
1 = Yes
2 = No
-9 = Not answered

[PK07] Horizontal Selection
Familiarity with Munich
"How familiar are you with the city of Munich and its surroundings?"

PK07 Familiarity with Munich
1 = Not at all familiar
2 = Slightly familiar
3 = Familiar
4 = Very familiar
5 = Extremely familiar
-9 = Not answered

[PK08] Selection
Rate expertise
"How would you rate your expertise in machine learning?"

PK08 Rate expertise
1 = Neophyte: Never used it in practice.
2 = Beginner: Basic understanding, some theoretical knowledge.
3 = Intermediate: Applied it in practice occasionally, familiar with common algorithms.
4 = Advanced: Regularly use it in practice, strong understanding of various techniques and tools.
5 = Expert: Extensive experience, deep understanding, and possibly contribute to the field through research or teaching.
-9 = Not answered

[PK09] Selection
Trained dataset
"Have you ever created a dataset to train a machine learning model?"

PK09 Trained dataset
1 = Yes
2 = No
-9 = Not answered

[PK10] Selection
Trained/tested
"Have you ever trained and tested a machine learning model on a dataset?"

PK10 Trained/tested
1 = Yes
2 = No
-9 = Not answered

Section PU: Perceived Understanding

[PU01] Horizontal Selection
Understand predictions
"After completing the experiment, I understand the reasoning behind the classifier's predictions."

PU01 Understand predictions
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[PU02] Horizontal Selection
Alignment with expectations
"The predictions of the classifier align with my understanding of machine learning image classification."

PU02 Alignment with expectations
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

Section ST: Strategies

[ST01] Selection
Approach to test set creation
"Which of the following best describes your strategy for creating your test set?"

ST01 Approach to test set creation
1 = I mainly selected challenging images for the classifier to uncover where the system fails.
2 = I mainly selected prototypical images for the classifier to assert where they system succeeds.
3 = I selected images that would produce a test set with equal number of correct and incorrect predictions.
4 = I selected images without a specific strategy.
-9 = Not answered

[ST02] Selection
Factors influencing prioritization
"Which of the following factors mostly influenced your choice of which classes to prioritize, as the testing ..."

ST02 Factors influencing prioritization
1 = The bar chart
2 = The outcome (pass/fail) in the table
3 = The confidence value in the table
4 = The predicted label in the table
5 = My intuition or judgement
6 = Other (please specify):
-9 = Not answered

ST02_06 Other (please specify)
Free text

[ST03] Selection
Surprising discrepancies
"Was there anything in the behavior of the classifier that you found surprising?"

ST03 Surprising discrepancies
1 = Yes (please explain what were your surprises):
2 = No
-9 = Not answered

ST03_01 Yes (please explain what were your surprises)
Free text

[ST04] Text Input
Information missing
"Is there any type of information that you feel would have been helpful in your testing, but was missing from..."

ST04_01 [01]
Free text

[ST05] Horizontal Selection
Usefulness of test coverage
"The bar chart showing the number of examples per class was useful to create my test set."

ST05 Usefulness of test coverage
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[ST06] Horizontal Selection
Usefulness of table info
"The table information (pass/fail outcome, predicted label, confidence) was useful to create my test set."

ST06 Usefulness of table info
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[ST07] Text Input
Strategy
"Please briefly explain your strategy for creating your test set, if any:"

ST07_01 [01]
Free text

Section TR: Trust and reliance

[TR01] Horizontal Selection
Reliable predictions
"I feel that the predictions provided by the satellite image classifier are reliable."

TR01 Reliable predictions
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR02] Horizontal Selection
Trust retrained model
"I would trust the retrained satellite image classifier more if it were updated based on the test set I created."

TR02 Trust retrained model
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR03] Horizontal Selection
Viewing disagreements
"Viewing the disagreements between my expected labels and the predicted labels of the classifier made me unde..."

TR03 Viewing disagreements
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR04] Horizontal Selection
Unpredictable reaction
"The system reacts unpredictably."

TR04 Unpredictable reaction
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR05] Horizontal Selection
Ability to understand
"I was able to understand why the image classifier made mistakes."

TR05 Ability to understand
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR06] Horizontal Selection
Real world app trust
"I would trust the system if it was deployed in a real-world application."

TR06 Real world app trust
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR07] Horizontal Selection
Understandable errors
"The system makes understandable errors."

TR07 Understandable errors
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR08] Horizontal Selection
Rely on the system
"I can rely on the system."

TR08 Rely on the system
1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered

[TR09] Horizontal Selection

Accurate classifier

"Overall, the image classifier is accurate."

TR09 Accurate classifier

1 = Strongly Disagree
2 = Disagree
3 = Neither Agree nor Disagree
4 = Agree
5 = Strongly Agree
-9 = Not answered